

Agentic AI Interview Process in Big Tech Companies

In a big tech company, an Agentic AI interview usually checks five main areas:

1. Coding
2. AI Agent Fundamentals
3. Live Agent Development
4. Agentic AI System Design
5. Production, Security, and Behavioral Skills

A candidate may face around **5 to 8 interview rounds**, depending on the company and role.

Step 1: Recruiter / Resume Screening

What they check

- Your background
- Your AI projects
- Your experience with Agentic AI
- Communication skill

Common Questions

- Tell me about yourself.
- Why do you want to work in Agentic AI?
- What AI agents have you built?
- Explain your most complex AI project.
- Did you deploy your project?

Candidate Preparation

Be ready to clearly explain:

- Project problem
- Architecture
- Technologies used
- Your contribution
- Challenges
- Final result

Step 2: Python and DSA Interview

Big tech companies may still ask coding and DSA questions.

Important Topics

- Array
- String
- HashMap
- Stack
- Queue
- Linked List
- Tree
- Graph
- BFS
- DFS
- Heap
- Binary Search
- Topological Sort
- Basic Dynamic Programming

Example Questions

- Implement an LRU Cache.
 - Detect a cycle in a graph.
 - Create a task scheduler.
 - Implement exponential backoff.
 - Process multiple API requests asynchronously.
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Step 3: Agentic AI Fundamentals

This round checks whether you understand AI agents properly.

Important Questions

- What is an AI Agent?
- AI Agent vs Chatbot?
- What is an Agent Loop?
- What is ReAct?
- What is Tool Calling?
- Function Calling vs MCP?
- What is Agent State?
- What is Agent Memory?
- What is Routing?
- What is Planning?
- Router Agent vs Supervisor Agent?
- Single Agent vs Multi-Agent?

Basic Agent Flow

User



LLM



Tool Selection



Tool Execution



Tool Result



LLM Decision



Final Answer

Important Interview Question

How do you stop an infinite agent loop?

Possible Solutions:

- Maximum iteration limit
- Recursion limit
- Timeout

- Token budget
 - Step limit
 - Repeated action detection
 - Proper END condition
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Step 4: Live Agent Coding

In this round, the interviewer may ask you to build an AI agent.

Example Task

Build a customer support agent with these tools:

```
tools = [  
    search_order,  
    cancel_order,  
    refund_order  
]
```

The agent should select the correct tool based on the user's request.

Then the interviewer may ask

- What if the API fails?
- What if the tool is slow?
- What if the tool returns invalid JSON?
- What if the same tool executes twice?
- What if the LLM selects the wrong tool?

Expected Solutions

API Failure

→ Retry

Slow Tool

→ Timeout

Invalid JSON

→ Schema Validation

Duplicate Execution

→ Idempotency

Wrong Tool Selection
→ Tool Validation

The interviewer checks your engineering thinking, not only your LangGraph syntax.

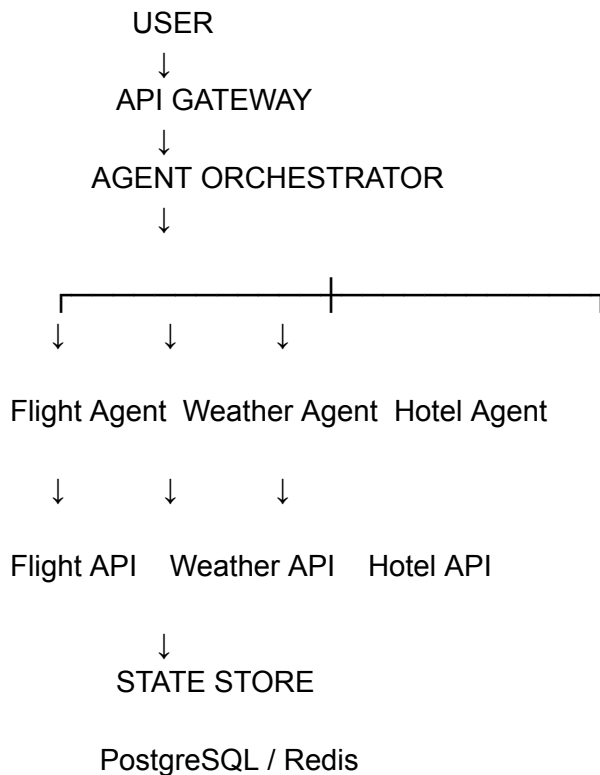
Step 5: Agentic AI System Design

This is one of the most important rounds.

Example Question

Design an AI Travel Agent.

Basic Architecture



Interview Questions

- Why did you use LangGraph?
- Why PostgreSQL?

- Why Redis?
 - What happens if the agent crashes?
 - How will you resume the workflow?
 - How will you handle one million users?
 - How will you reduce latency?
 - How will you reduce token cost?
 - How do agents communicate?
 - How do you prevent duplicate tool calls?
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Step 6: Agent Evaluation

You must know how to measure agent performance.

Important Metrics

- Task Success Rate
- Tool Selection Accuracy
- Tool Argument Accuracy
- Hallucination Rate
- Latency
- Token Usage
- Cost Per Task
- Recovery Rate
- Human Escalation Rate
- Safety Violation Rate

Example

1000 Agent Tests

850 → Success

50 → Wrong Tool

30 → Invalid Arguments

25 → Hallucination

20 → Tool Timeout

15 → Infinite Loop

10 → Security Failure

The goal is to identify the failure category and improve the agent.

Step 7: Agent Security

AI Agents can access tools and external systems.

Because of this, security is very important.

Common Questions

- What is Prompt Injection?
- What is Indirect Prompt Injection?
- How do you secure MCP tools?
- Should an LLM directly access a database?
- How do you protect API keys?
- How do you prevent dangerous tool execution?

Important Security Concepts

- Least Privilege
- Tool Allowlist
- Input Validation
- Output Validation
- Structured Schema
- Permission Layer
- Human Approval
- Audit Logs
- Secret Management

Unsafe Architecture

Agent
↓
Delete Database

Better Architecture

Agent
↓
Permission Layer
↓
Policy Validation
↓
Human Approval
↓
Execute Action

Step 8: AI and LLM Fundamentals

You should also prepare basic LLM concepts.

Important Topics

- Transformer
- Attention
- Self-Attention
- Embeddings
- Cosine Similarity
- Temperature
- Top-P
- Context Window
- KV Cache
- Quantization
- Fine-Tuning
- RAG
- LoRA
- RLHF
- Hallucination
- Structured Output

Step 9: Behavioral Interview

The interviewer checks how you solve real problems.

Common Questions

- Tell me about a failed project.
 - Tell me about a production issue.
 - Tell me about a technical disagreement.
 - Tell me about a difficult problem.
 - Tell me about a wrong technical decision.
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Typical Agentic AI Interview Rounds

Round	Interview Type
1	Recruiter / Resume
2	Python + DSA
3	Agentic AI Fundamentals
4	Live Agent Coding
5	Agentic AI System Design
6	Evaluation + Security
7	Behavioral / Hiring Manager

Recommended Preparation

30% → Python and DSA

25% → Agentic AI Fundamentals

25% → Agentic AI System Design

10% → Evaluation and Security

10% → Behavioral Interview
